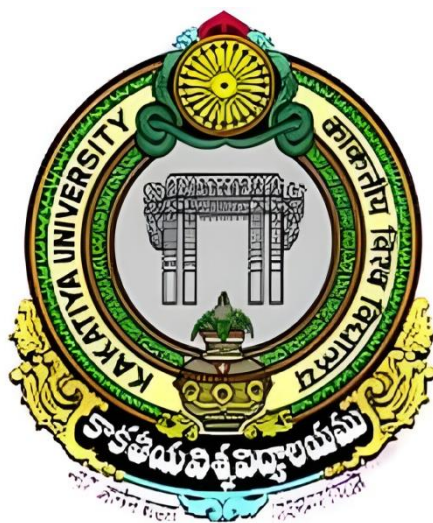


B.Sc. PHYSICS
SYLLABUS
(Common core syllabus under CBCS)

With effect from the academic year: 2025-2026 and onwards



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DEPARTMENT OF PHYSICS
KAKATIYA UNIVERSITY
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Annexure-I (Credits)

Proposed CBCS Structure from 2025-2026 for Under Graduate Course

Courses		Papers	Total Credits	Credits for each paper/ Semester					
				B.Sc					
				I	II	III	IV	V	VI
Core Courses (DSC)	Major-1	6	30	5	5	5	5	5	5
	Major-2	6	30	5	5	5	5	5	5
	Minor-1	4	20	5	5	5	5	---	---
MIL/AEC (First language)	English	4	20	5	5	5	5	---	---
Second Language (Telugu, Hindi, Urdu etc.,)		4	20	5	5	5	5	---	---
Multi Disciplinary Course	MDC-1	1	4	---	---	---	---	4	---
SEC 1,2		2	4	---	---	---	---	2	2
SEC 3,4		2	4	---	---	---	---	2	2
Value added course (VAC)	VAC 1,2	2	6	---	---	---	---	3	3
Internships	Internship/Project	1	4	---	---	---	---	---	4
Total Credits in each semester		---	142	25	25	25	25	21	21
Total Credits in UG		---	142						


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B.Sc. PHYSICS SYLLABUS

SCHEME OF INSTRUCTIONS UNDER CBCS (w.e.f. 2025-26 academic year onwards)

Year	Semester	Title of the Paper [Theory and Practical]	Instructions Hrs/week	Number of Credits	Total Credits	Marks	
1st Year	I Sem	Paper – I: Mechanics and Oscillations	4	4	5	100	
		Practicals – I: Mechanics and Oscillations Lab	2	1		25	
	II Sem	Paper – II: Thermal Physics	4	4	5	100	
		Practicals – II: Thermal Physics lab	2	1		25	
2nd Year	III Sem	Paper – III: Electromagnetic Theory	4	4	5	100	
		Practicals – III : Electromagnetic Theory Lab	2	1		25	
	IV Sem	Paper – IV : Optics	4	4	5	100	
		Practicals – IV : Optics lab	2	1		25	
3rd Year	V Sem	Paper –V : Modern Physics	4	4	5	100	
		Practicals – V: Modern Physics lab	2	1		25	
		Multi Disciplinary Courses (MDC-1):					
		Radiation Physics	4	4	4	100	
		Skill Enhancement Courses (SEC):					
		<u>SEC-1:</u> Communications Skills/Professional Development Skills/ Entrepreneurship & Starts up	2	2	2	50	
		<u>SEC-3:</u> Fundamentals of AI Tools/Ability Skills (Competitive Mathematics)	2	2	2	50	
		Value Added Course (VAC)					
		<u>VAC-1-Paper-1:</u> Environmental Science (EVS)/ Cyber Security & Cyber laws	3	3	3	75	
	VI Sem	Paper – VI : Solid State Physics & Solid State Devices	4	4	5	100	

		Practicals VI : Solid State Physics & Solid State Devices lab	2	1		25
		Skill Enhancement Courses (SEC):				
		<u>SEC-2:</u> Professional Development Skills /Communications Skills/Entrepreneurship & Starts up	2	2	2	50
		<u>SEC-4:</u> Biomedical instrumentation	2	2	2	50
		Value Added Course (VAC)				
		<u>VAC-2--Paper-2:</u> Cyber Security & Cyber laws/Environmental Science (EVS)	3	3	3	75
		Project work /Internship:				
		(Innovative Products making Skill (IPMS))	4	4	4	100

Total Credits: 52


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KAKATIYA UNIVERSITY-WARANGAL-TELANGANA

Under Graduate Courses (Under CBCS 2025–2026 onwards)

B.Sc (PHYSICS) - I Year, SEMESTER – I

Paper–I: Mechanics and Oscillations

w.e.f academic year (2025-26) (CBCS)

Total: 56 Hrs (4hrs/week)

UNIT–I

Vector Analysis: (7 hrs)

Scalar and Vector fields, Gradient of a Scalar field, Divergence and Curl of a Vector field and their physical significance and related problems. Vector integration, Line, Surface and Volume integrals. Applications of Stokes', Gauss's and Green's theorems.

Rigid body Dynamics: (7 hrs)

Definition of Rigid body, rotational kinematic relations, equation of motion for a rotating body, angular momentum and inertial tensor. Euler's equations, precession of a top, Gyroscope.

UNIT–II

Central Forces: (7 hrs)

Central forces-definition and examples, conservative nature of central forces, conservative force as a negative gradient of potential energy, equation of motion under a central force, gravitational potential and gravitational field, motion under inverse square law, derivation of Kepler's laws.

Special theory of Relativity: (7 hrs)

Galilean relativity, absolute frames, Michelson-Morley experiment, Postulates of special theory of relativity. Lorentz transformation equations, time dilation, length contraction, addition of velocities, mass-energy relation, Concept of four vector formalism.

UNIT – III

Oscillations: (14 hrs)

Simple harmonic oscillator and solution of the differential equation-Physical characteristics of

SHM, torsion pendulum measurements of rigidity modulus, compound pendulum, measurement of 'g', Damped harmonic oscillator, solution of the differential equation of damped oscillator, Energy considerations, logarithmic decrement, relaxation time, quality factor, differential equation of forced oscillator and its solution, amplitude resonance, velocity resonance.

UNIT–IV

Waves: (14 hrs)

Fundamentals of Waves -Transverse wave propagation along a stretched string, general solution of wave equation and its significance, modes of vibration of stretched string clamped at ends, overtones, energy transport, transverse impedance.

Longitudinal vibrations in bars- wave equation and its general solution,

Special cases: i) Bar fixed at both ends, ii) Bar fixed at the midpoint, iii)

Bar free at both ends, iv) Bar fixed at one end, Transverse vibrations in a

bar - wave equation and its general solution.


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Note: Problems should be solved at the end of every chapter of all units.

Reference/Suggested Books

- 1) **Mechanics** by C.Kittel,W.Knight,M.A.Ruderman-Berkeley Physics Course.Vol.1, *Tata-Mc Grawhill Company Edition 2008.*
- 2) **Fundamentals of Physics.**Halliday/Resnick/Walker *Wiley India Edition 2007.*
- 3) **Theory of relativity - Resnick**
- 4) **First Year Physics-Telugu Academy, Telangana**
- 5) **Introduction to Physics for Scientists and Engineers.** F.J. Ruche. *McGraw Hill.*
- 6) **Fundamentals of Physics** by Alan Giambattista et al *Tata-Mc Graw Hill Company*
Edition, 2008.
- 7) **University Physics** by Young and Freeman, *Pearson Education, Edition 2005.*
- 8) **Sears and Zemansky's University Physics** by Hugh D.Young, Roger A.Freedman
Pearson Education Eleventh Edition.
- 9) **An introduction to Mechanics** by Daniel Kleppner & Robert Kolenkow. *The McGraw Hill Companies.*
- 10) **Mechanics** by Hans & Puri. *TMH Publications.*
- 11) **Engineering Physics.** R.K.Gaur & S.L. Gupta. *Dhanpat Rai Publications.*
- 12) **The Feynman Lectures in Physics, Vol.-1,** R P Feynman, R B Lighton and M Sands,
BI Publications,
- 13) **Mechanics** by P.K.Srivastava-NewAge International.
- 14) **Mathematical Physics** by Satya Prakash- Sultan Chand & Sons.
- 15) **Vector Analysis** by Murray R.Spiegel-2nd edition-Schaum's Outlines,Mc GrawHill
Education


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B.Sc (Physics) -I year, Semester - I
Paper-I: Mechanics & Oscillations Practical lab

No. of hours per week: 2

- 1) Simple pendulum – Measurement of errors
- 2) Study of a compound pendulum-determination of ‘g’ and ‘k’.
- 3) Determination of Young modulus by uniform bending of a wooden/ metal bar method.
- 4) Determination of moment of inertia of a flywheel.
- 5) Determination of rigidity modulus by torsion pendulum.
- 6) Determine of Viscosity of a fluid by poissuelle method.
- 7) Determination of oscillations of a given spring constant and frequency by using combination of springs-series and parallel.
- 8) Study of Oscillations under bifilar suspension-Verification of axis theorems.
- 9) Determine surface tension of a liquid through capillary rise method.
- 10) Determine surface tension of a liquid by any other method.
- 11) Verification of laws of a stretched string using Sonometer. (Three Laws).
- 12) Calculation of slope and intercept of a $Y = mX + C$ graph by theoretical method (simple pendulum experiment)
- 13) Determination of frequency of a Bar-Melde’s experiment
- 14) Experimental analysis of gyroscope using simulation.
- 15) Verification of Stokes, Gauss-Divergence and Green’s theorem using simulation.

Note: Minimum of **Eight** experiments should be performed. Maximum of 15 students per batch and maximum of three students per experiment should be allotted in the regular practical class of three hours per week.

Suggested Books

- 1) D.P.Khandelwal, “A laboratory manual for under graduate classes” (Vani Publishing House, New Delhi).
- 2) S.P.Singh, “Advanced Practical Physics” (Pragati Prakashan, Meerut).
- 3) Worsnop and Flint-Advanced Practical physics for students.
- 4) “Practical Physics” R.KShukla, Anchal Srivastava.


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KAKATIYA UNIVERSITY-WARANGAL-TELANGANA
Under Graduate Courses (Under CBCS 2025–2026 onwards)

B.Sc (PHYSICS) I Year, SEMESTER – II

Paper-II: Thermal Physics

w.e.f academic year (2025-26) (CBCS)

Total: 56 Hrs (4hrs/week)

UNIT–I

Kinetic theory of gases: (4 hrs)

Introduction-Deduction of Maxwell's law of distribution of molecular speeds, Transport

Phenomena-Viscosity of gases-thermal conductivity – diffusion of gases.

Thermodynamics: (8 hrs)

Basics of Thermodynamics-Carnot's engine (qualitative)-Carnot's theorem-Kelvin's and

Clausius statements-Thermodynamic scale of temperature-Entropy, physical significance-

Change in entropy in reversible and irreversible processes-Entropy and disorder-Entropy of

universe -Temperature-Entropy (T-S) diagram - Change of entropy of a perfect gas-change of

entropy when ice changes into steam, Application of entropy in waste management.

UNIT–II

Thermodynamic potentials and Maxwell's equations: (8 hrs)

Thermodynamic potentials-Derivation of Maxwell's thermodynamic relations-Clausius-

Clayperon's equation-Derivation for ratio of specific heats-Derivation for difference of two specific heats for perfect gas.

Low temperature Physics: (8 hrs)

Joule Kelvin effect-liquefaction of gas using porous plug experiment, Joule expansion-Distinction between adiabatic and Joule Thomson expansion-Expression for Joule Thomson cooling-Liquefaction of helium, Kapitza's method-Adiabatic demagnetization-Production of low temperatures -Principle of refrigeration, vapour compression type, Thermocouple-seebeck effect, Peltier effect and Thomson's effect.

UNIT–III

Quantum theory of radiation: (14 hrs)

Black body-Ferry's black body-distribution of energy in the spectrum of Black body-

Wein's displacement law, Wein's law, Rayleigh-Jean's law-Quantum theory of radiation-Planck's law-deduction of Wein's law, Rayleigh-Jeans law, Stefan's law from Planck's law.

Measurement of radiation using pyrometers-Disappearing filament optical pyrometer-experimental determination-Angstrom pyro heliometers-determination of solar constant, effective temperature of sun.

UNIT-IV

Statistical Mechanics: (14 hrs)

Introduction, postulates of statistical mechanics. Phase space, concept of ensembles and some known ensembles, classical and quantum statistics and their differences, concept of probability, Maxwell-Boltzmann's distribution law - Molecular energies in an ideal gas - Maxwell-Boltzmann's velocity distribution law (qualitative), Bose-Einstein Distribution law- application to Photon energy, Fermi-Dirac Distribution law- free electron gas, comparison of three distribution laws.

NOTE: Problems should be solved at the end of every chapter of all units.

Reference/Suggested books

- 1) **Fundamentals of Physics**.byHalliday/Resnick/Walker.C.Wiley India Edition 2007.
- 2) **Second Year Physics** –Telugu Academy, Telangana
- 3) **Modern Physics** by R.Murugeshan and Kiruthiga Siva Prasath (For Statistical mechanics) S. Chand & Co.
- 4) **Modern Physics** by G.Aruldas and P. Rajagopal, Eastern Economy Education.
- 5) **Statistical Physics** by F.Reif Berkeley Physics Course.Volume-5,The McGraw-Hill Companies.
- 6) **An Introduction to Thermal Physics** by Daniel V. Schroeder.Pearson Education Low Price Edition.
- 7) **Thermodynamics** by R.C. Srivastava, Subit K. Saha & Abhay K. Jain Eastern Economy Edition.
- 8) **Modern Engineering Physics** by A.S.Vasudeva .S.Chand & Co. Publications.
- 9) **Feynman's Lectures on Physics** Vol. 1, 2, 3 & 4.Narosa Publications.
- 10) **Heat and Thermo dynamics:** K.W.Zeemansky.
- 11) **Introduction to statistical Mechanics**”by B.B. Laud (Macmillan 1981).
- 12) **Statistical Physics**” by F.Reif., (Mc Graw-Hill,1998)
- 13) **Statistical Physics**” by K.Haung., (Wiley Eastern 1988)


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B.Sc (Physics) - I year, Semester - II
Paper-II: Thermal Physics Practical lab

No. of hours per week: 2

- 1) Determination of Co-efficient of thermal conductivity of a bad conductor by Lee's method.
- 2) Determination of Stefan's constant-Stefan's experiment.
- 3) Determination of Specific heat of a liquid by using Newton's law of cooling method.
- 4) Determination of heating efficiency of electrical kettle with varying voltages.
- 5) Cooling Curve of a metallic body (Null method).
- 6) Determination of temperature coefficient of resistance using resistance thermometer.
- 7) Study of conversion of mechanical energy to heat.
- 8) Determination of Specific heat of a solid (graphite).
- 9) Thermal expansion of solids
- 10) Calibration of thermo couple
- 11) Simulations for T-S diagram

Note: Minimum of **Eight** experiments should be performed. Maximum of 15 students per batch and maximum of three students per experiment should be allotted in the regular practical class of three hours per week.

Suggested Books

- 1) D.P.Khandelwal, "A laboratory manual for under graduate classes"(Vani Publishing House, New Delhi).
- 2) S.P.Singh, "Advanced Practical Physics"(Pragati Prakashan, Meerut).
- 3) Worsnop and Flint-Advanced Practical physics for students.
- 4) "Practical Physics" R.KShukla, Anchal Srivastava.


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Internal Question paper pattern
Faculty of Science
B.Sc Physics

Duration: 90 Minutes]

[Max. Marks: 20

Semester:

Internal: I

Subject:

Date:

Paper:

Answer all the Questions

Each question carries equal marks (2 x 10 = 20)

- 1) Unit – I
- 2) Unit – I
- 3) Unit – I
- 4) Unit – I
- 5) Unit – I
- 6) Unit – II
- 7) Unit – II
- 8) Unit – II
- 9) Unit – II
- 10) Unit – II


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Internal Question paper pattern

Faculty of Science

B.Sc Physics

Duration: 90 Minutes]

[Max. Marks: 20

Semester:

Internal: II

Subject:

Date:

Paper:

Answer all the Questions

Each question carries equal marks (2 x 10 = 20)

- 1) Unit – III
- 2) Unit – III
- 3) Unit – III
- 4) Unit – III
- 5) Unit – III
- 6) Unit – IV
- 7) Unit – IV
- 8) Unit – IV
- 9) Unit – IV
- 10) Unit – IV


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Question paper pattern

Faculty of Science

B.Sc Physics

Title of the paper:

Paper:

Duration: 3Hrs]

[Max. Marks: 80

Section-A: Short Answer Questions

(8 x 4 = 32)

Answer any EIGHT questions

- 1) Unit – I
- 2) Unit – I
- 3) Unit – I (Problem)
- 4) Unit – II
- 5) Unit – II
- 6) Unit – II (Problem)
- 7) Unit – III
- 8) Unit – III
- 9) Unit – III (Problem)
- 10) Unit – IV
- 11) Unit – IV
- 12) Unit – IV (Problem)

Section B: Essay Answer Questions

(4 x 12 = 48)

- 13) (a) Unit – I
OR

(b) Unit – I

- 14) (a) Unit – II

OR

(b) Unit – II

15) (a) Unit – III

OR

(b) Unit – III

16) (a) Unit – IV

OR

(b) Unit – IV


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Annexure – I (Credits)

Proposed CBCS Structure from 2025-26 for Under Graduate Courses

Courses		Paper s	Total Credits	Credits for each paper / Semester						Credits for each paper / Semester						Credits for each paper / Semester					
				BA						B.Com.						B.Sc.					
				I	II	III	IV	V	VI	I	II	III	IV	V	VI	I	II	III	IV	V	VI
Core Courses DSC	Major-1	6	30	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
	Major - 2	6	30	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
	Minor-1	4	20	5	5	5	5	-	-	5	5	5	5	-	-	5	5	5	5	-	-
MIL/AEC (First Language)	English	4	20	5	5	5	5	-	-	5	5	5	5	-	-	5	5	5	5	-	-
Second Language (Telugu, Hindi, Urdu, etc.)		4	20	5	5	5	5	-	-	5	5	5	5	-	-	5	5	5	5	-	-
Multi-Disciplinary Course	MDC 1	1	4	-	-	-	-	4	-	-	-	-	-	4	-	-	-	-	-	4	-
Sec 1, 2		2	4					2	2					2	2					2	2
Sec 3, 4		2	4					2	2					2	2					2	2
Value added course (VAC)	VAC 1, 2	2	6	-	-	-	-	3	3	-	-	-	-	3	3	-	-	-	-	3	3
Internships	Internship / Project	1	4	-	-	-	-	-	4	-	-	-	-	-	4	-	-	-	-	-	4
Total Credits in each semester			142	25	25	25	25	21	21	25	25	25	25	21	21	25	25	25	25	21	21
Total Credits in UG				142						142						142					
Credits under Non-CGPA (Community engagement and service)		NSS /NCC /sports / Extra curricular	6	Upto 6 (2 in each year)						Upto 6 (2 in each year)						Upto 6 (2 in each year)					
		IKS	4	Upto 4 (2 in each, after I & II years)						Upto 4 (2 in each, after I & II years)						Upto 4 (2 in each, after I & II years)					

**B.Sc. PHYSICS
SYLLABUS
(Common core syllabus under CBCS)**

With effect from the academic year: 2025-2026 and onwards



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Annexure-I (Credits)

Proposed CBCS Structure from 2025-2026 for Under Graduate Course

Courses		Papers	Total Credits	Credits for each paper/ Semester					
				B.Sc					
				I	II	III	IV	V	VI
Core Courses (DSC)	Major-1	6	30	5	5	5	5	5	5
	Major-2	6	30	5	5	5	5	5	5
	Minor-1	4	20	5	5	5	5	---	----
MIL/AEC (First language)	English	4	20	5	5	5	5	---	----
Second Language (Telugu, Hindi, Urdu etc.,)		4	20	5	5	5	5	---	----
Multi Disciplinary Course	MDC-1	1	4	---	----	---	----	4	----
SEC 1,2		2	4	---	----	---	----	2	2
SEC 3,4		2	4	---	----	---	----	2	2
Value added course (VAC)	VAC 1,2	2	6	---	----	---	----	3	3
Internships	Internship/Project	1	4	---	----	---	----	---	4
Total Credits in each semester		----	142	25	25	25	25	21	21
Total Credits in UG		---	142						


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B.Sc. PHYSICS SYLLABUS

SCHEME OF INSTRUCTIONS UNDER CBCS (w.e.f. 2025-26 academic year onwards)

Year	Semester	Title of the Paper [Theory and Practical]	Instructions Hrs/week	Number of Credits	Total Credits	Marks	
1st Year	I Sem	Paper – I: Mechanics and Oscillations	4	4	5	100	
		Practicals – I: Mechanics and Oscillations Lab	2	1		25	
	II Sem	Paper – II: Thermal Physics	4	4	5	100	
		Practicals – II: Thermal Physics lab	2	1		25	
2nd Year	III Sem	Paper – III: Electromagnetic Theory	4	4	5	100	
		Practicals – III : Electromagnetic Theory Lab	2	1		25	
	IV Sem	Paper – IV : Optics	4	4	5	100	
		Practicals – IV : Optics lab	2	1		25	
3rd Year	V Sem	Paper –V : Modern Physics	4	4	5	100	
		Practicals – V: Modern Physics lab	2	1		25	
		Multi Disciplinary Courses (MDC-1):					
		Radiation Physics		4	4	4	100
		Skill Enhancement Courses (SEC):					
		<u>SEC-1:</u> Communications Skills/Professional Development Skills/ Entrepreneurship & Starts up		2	2	2	50
		<u>SEC-3:</u> Fundamentals of AI Tools/Ability Skills (Competitive Mathematics)		2	2	2	50
		Value Added Course (VAC)					
		<u>VAC-1-Paper-1:</u> Environmental Science (EVS)/ Cyber Security & Cyber laws		3	3	3	75

VI Sem	Paper – VI : Solid State Physics & Solid State Devices	4	4	5	100
	Practicals VI : Solid State Physics & Solid State Devices lab	2	1		25
	Skill Enhancement Courses (SEC):				
	<u>SEC-2:</u> Professional Development Skills /Communications Skills/Entrepreneurship & Starts up	2	2	2	50
	<u>SEC-4:</u> Biomedical instrumentation	2	2	2	50
	Value Added Course (VAC)				
	<u>VAC-2--Paper-2:</u> Cyber Security & Cyber laws/Environmental Science (EVS)	3	3	3	75
	Project work /Internship:				
	(Innovative Products making Skill (IPMS))	4	4	4	100


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Under Graduate Courses (Under CBCS 2025–2026 onwards)

B.Sc (PHYSICS) II Year, SEMESTER-III **PAPER-III: ELECTROMAGNETIC THEORY** w.e.f academic year (2025-26) (CBCS)

Total: 56 Hrs (4hrs/week)

UNIT-I

Electrostatics: (8 hrs)

Electric Field, Concept of electric field lines and electric flux, Gauss's law (Integral and differential forms), application of Gauss's law-linear and plane, spherical charge distributions, Conservative nature of electric field 'E', Irrotational field, Electric potential: Concept of electric potential, relation between electric potential and electric field, potential energy of a system of charges, Energy density in an electric field, Calculation of potential from electric field for a spherical charge distribution.

Dielectrics: (6 hrs)

Dielectric properties of matter, Electric field in matter, polarization, polarization charge, electric susceptibility & dielectric constant, capacitors (Parallel, Spherical, Cylindrical plates) filled with dielectrics, Displacement vector D, Gauss Law in dielectrics, Relation of E, P & D.

UNIT-II

Magnetostatics: (14 hrs)

Concept of magnetic field 'B' and magnetic flux, Biot-Savart's law, B' due to a straight current carrying conductor, Force on a point charge in a magnetic field, properties of magnetic induction (B), curl and divergence of B, solenoidal field, Integral form of Ampere's law, Applications of Ampere's law- field due to current carrying straight conductor, circular coil and solenoidal, Energy stored in magnetic field., Magnetic energy in terms of current and inductance, Magnetic force between two current carrying conductors, Magnetic field intensity, Ballistic Galvanometer, Torque on a current loop in a uniform magnetic field, working principle of Ballistic Galvanometer, current and charge sensitivity, electromagnetic damping, critical damping resistance.

UNIT-III:

Electromagnetic Induction and Electromagnetic waves: (14 hrs)

Faraday's laws of induction (differential and integral form), Lenz's law, self and mutual Induction, Continuity equation, modification of Ampere's law, displacement current, Maxwell equations, Maxwell's equations in vacuum and dielectric medium, boundary conditions, Poynting's theorem, plane wave equation, transverse nature of EM waves, velocity of light in vacuum and in medium.

UNIT-IV:

Varying and alternating currents: (8 hrs)

Introduction, time constants in a LR and CR circuits (Growth and decay of currents), Growth and decay of currents in LCR circuits - Critical damping, Alternating current, relation between current and voltage in pure R,C and L-vector diagrams - Power in ac circuits, LCR series and parallel resonant circuit, frequency,Q-factor, AC & DC motors-single phase, three phase (basics only). RC low pass, High pass filters.



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Network Theorems (6 hrs)

Passive elements, power sources, Active elements, Network models: T and π Transformations, Superposition theorem, Thevenin's theorem, Norton's theorem. Reciprocity theorem and Maximum power transfer theorem (problems).

NOTE: Problems should be solved at the end of every chapter of all units.

Reference/Suggested Books:

- 1) **Fundamentals of electricity and magnetism** by Arthur F.Kip (McGraw-Hill,1968)
- 2) **Electricity and magnetism** by J.H.Fewkes & JohnYarwood.Vol.I (Oxford Univ.Press, 1991).
- 3) **Introduction to Electrodynamics**,3rd edition, by DavidJ.Griffiths,(Benjamin Cummings,1998).
- 4)**Electricity and magnetism** by Edward M.Purcell (McGraw-HillEducation,1986)
- 5)**Electricity and magnetism**.by DC Tayal (Himalaya Publishing House,1988)
- 6) **Electromagnetics** by Joseph A.Edminister 2nd ed., (NewDelhi:TataMcGrawHill, 2006).
- 7) **II year Physics** by Telugu Academy, Telangana



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B.Sc (Physics) -II year, Semester – III
Paper-III: Electromagnetic Theory practical lab

No. of hours per week: 2

- 1) To verify the Thevenin's Theorem
- 2) To verify Norton's Theorem
- 3) To verify Superposition Theorem
- 4) To verify maximum power transfer theorem
- 5) To determine a small resistance by Carey Foster's bridge.
- 6) To determine the (a) current sensitivity, (b) charge sensitivity, and (c) CDR of a B.G.
- 7) To determine high resistance by leakage method.
- 8) To determine coefficient of Mutual inductance by absolute method.
- 9) Determine the Band width, quality factor and resonance frequency of a Parallel LCR Circuit.
- 10) To determine self inductance of coil by Anderson's bridge using AC
- 11) Determine the Band width, quality factor and resonance frequency of a Series LCR Circuit.
- 12) Determine the time constant of L-R circuit.
- 13) Determine the time constant of RC circuit.
- 14) Determination of internal resistance of a cell by using potentiometer.
- 15) To determine the ratio of two capacitances by De Sauty's bridge.

Note: Minimum of **Eight** experiments should be performed. Maximum of 15 students per batch and maximum of three students per experiment should be allotted in the regular practical class of three hours per week.

Suggested Books:

- 1) B.L.Worsnop and H.T.Flint Advanced Practical Physics, Asia Publishing House, New Delhi
- 2) InduPrakash and Ramakrishna, A Text Book of Practical Physics, KitabMahal


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KAKATIYA UNIVERSITY-WARANGAL-TELANGANA
Under Graduate Courses (Under CBCS 2025–2026 onwards)

B.Sc PHYSICS II Year, SEMESTER – IV

PAPER–IV: OPTICS

w.e.f academic year (2025-26) (CBCS)

Total: 56 Hrs (4hrs/week)

UNIT-I:

Interference: (2 hrs)

Principle of super position-coherence-temporal coherence and spatial coherence-conditions for Interference of light.

Interference by division of wave front: (5 hrs)

Fresnel's biprism-determination of wavelength of light, Determination of thickness of a transparent material using Fresnel's biprism-change of phase on reflection-Lloyd's mirror experiment.

Interference by division of amplitude: (7 hrs)

Oblique incidence of a plane wave on a thin film due to reflected and transmitted light (Cosine law), Colors of thin films - Non-reflecting films, Interference by a plane parallel film illuminated by a point source, Determination of diameter of wire using wedge shaped. Determination of wave length of monochromatic light – Michelson Interferometer-types of fringes-Determination of wavelength of monochromatic light using Newton's rings (both transmitted & reflected light).

UNIT-II

Diffraction: (14 hrs)

Introduction-Distinction between Fresnel and Fraunhofer diffraction, Fraunhofer diffraction-Diffraction due to single slit and circular aperture-Limit of resolution- Fraunhofer diffraction due to double slit-Fraunhofer diffraction pattern with N slits (diffraction grating). Resolving Power of grating-Determination of wave length of light in normal and oblique incidence methods using diffraction grating.

Fresnel diffraction-Fresnel's half period zones-area of the half period zones -zone plate-

Comparison of zone plate with convex lens-Phase reversal zone plate-diffraction at a straight edge difference between interference and diffraction.

UNIT III:

Polarization: (14 hrs)

Introduction, Methods of Polarization, Polarization by reflection, refraction, double refraction, selective absorption, scattering of light-Brewster's law-Malus law-Nicol prism, polarizer and analyzer-Refraction of plane wave incident on negative and positive crystals (Huygen's explanation) - Quarter wave plate, Half wave plate-Babinet's compensator, optical activity, analysis of light by Laurent's half shade polarimeter.

UNIT IV:

Lasers and Holography: (14 hrs)

Lasers: Introduction, Laser principle, Metastable states, Stimulated emission, spontaneous emission, Optical pumping, population inversion and, Three-Level and Four-Level Lasers. Construction and working of Semiconductor Laser, Relationship among Einstein coefficients, Applications of Lasers.


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Holography: Basic Principles of Holography- Recording of amplitude and phase- The recording medium- Reconstruction of original wave front- Image formation by wave front reconstruction. Gaber Hologram- Limitations of Gaber Hologram-Off axis Hologram- Fourier transforms Holograms- Volume Holograms, Applications of Holograms.

NOTE: Problems should be solved at the end of every chapter of all units

Reference/Suggested books:

- 1) **Optics** by Ajoy Ghatak. *The McGraw-Hill companies.*
- 2) **Optics** by Subramanian and Brijlal. *S.Chand & Co.*
- 3) **Fundamentals of Physics** by Halliday/Resnick/Walker. *C.Wiley India Edition 2007.*
- 4) **Optics and Spectroscopy** by R.Murugesan and Kiruthiga Siva Prasath. *S.Chand & Co.*
- 5) **Second Year Physics–Telugu Academy, Telangana**
- 6) **Modern Engineering Physics** by A.S. Vasudeva. *S.Chand & Co. Publications.*
- 7) **Feynman's Lectures on Physics** Vol.1, 2, 3 & 4. *Narosa Publications.*
- 8) **Fundamentals of Optics** by Jenkins A.Francis and White E.Harvey, *McGrawHill Inc.*
- 9) **Physical Optics** K.Ghatak,
- 10) **Optical and Atomic Physics** by D.P.Khandelwal, (Himalaya Publishing House, Bombay, 1988)
- 11) **Fundamental of Optics** by Jenkins and White: (McGraw-Hill)
- 12) **Optics** by Smith and Thomson: (John Wiley and sons).


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B.Sc (Physics)-II year, Semester – IV
Paper–IV: Optics practical lab

No. of hours per week: 2

- 1) Determination of thickness of a wire using wedge method.
- 2) Determination of wave length of light using Biprism.
- 3) Determination of Radius of curvature of a given convex lens by forming Newton's rings.
- 4) Resolving power of grating.
- 5) Study of optical rotation-polarimeter.
- 6) Dispersive power of a prism.
- 7) Determination of wavelength of light using diffraction grating minimum deviation method.
- 8) Wavelength of light using diffraction grating-normal incidence method.
- 9) Resolving power of a telescope.
- 10) Wavelength of Laser light using diffraction grating.
- 11) Refractive index of a liquid and glass (Boys Method).
- 12) Pulfrich refractometer – determination of refractive index of liquid.
- 13) To determine the wavelength of laser source using diffraction of single slit.
- 14) To determine the wavelength of laser source using diffraction of double slits

Note: Minimum of Eight experiments should be performed. Maximum of 15 students per batch and maximum of three students per experiment should be allotted in the regular practical class of three hours per week.

Suggested Books

1. D.P. Khandelwal, "A laboratory manual for undergraduate classes" (Vani Publishing House, New Delhi).
2. S.P. Singh, "Advanced Practical Physics" (Pragati Prakashan, Meerut).
3. Worsnop and Flint- Advanced Practical physics for students.
4. "Practical Physics" R.K Shukla, Anchal Srivastav


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Internal Question paper pattern
Faculty of Science
B.Sc Physics

Duration: 90 Minutes]

[Max. Marks: 20

Semester:

Internal: I

Subject:

Date:

Paper:

Answer all the Questions

Each question carries equal marks (2 x 10 = 20)

- 1) Unit – I
- 2) Unit – I
- 3) Unit – I
- 4) Unit – I
- 5) Unit – I
- 6) Unit – II
- 7) Unit – II
- 8) Unit – II
- 9) Unit – II
- 10) Unit – II


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Internal Question paper pattern
Faculty of Science
B.Sc Physics

Duration: 90 Minutes]

[Max. Marks: 20

Semester:

Internal: II

Subject:

Date:

Paper:

Answer all the Questions

Each question carries equal marks (2 x 10 = 20)

- 1) Unit – III
- 2) Unit – III
- 3) Unit – III
- 4) Unit – III
- 5) Unit – III
- 6) Unit – IV
- 7) Unit – IV
- 8) Unit – IV
- 9) Unit – IV
- 10) Unit – IV


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Question paper pattern

Faculty of Science

B.Sc Physics

Title of the paper:

Paper:

Duration: 3Hrs]

[Max. Marks: 80

Section-A: Short Answer Questions

(8 x 4 = 32)

Answer any EIGHT questions

- 1) Unit – I
- 2) Unit – I
- 3) Unit – I (Problem)
- 4) Unit – II
- 5) Unit – II
- 6) Unit – II (Problem)
- 7) Unit – III
- 8) Unit – III
- 9) Unit – III (Problem)
- 10) Unit – IV
- 11) Unit – IV
- 12) Unit – IV (Problem)

Section B: Essay Answer Questions

(4 x 12 = 48)

13) (a) Unit – I
OR

(b) Unit – I

14) (a) Unit – II
OR

(b) Unit – II

15) (a) Unit – III
OR

(b) Unit – III

16) (a) Unit – IV
OR

(b) Unit – IV


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Annexure – I (Credits)

Proposed CBCS Structure from 2025-26 for Under Graduate Courses

Courses		Paper s	Total Credits	Credits for each paper / Semester						Credits for each paper / Semester						Credits for each paper / Semester					
				BA						B.Com.						B.Sc.					
				I	II	III	IV	V	VI	I	II	III	IV	V	VI	I	II	III	IV	V	VI
Core Courses DSC	Major-1	6	30	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
	Major - 2	6	30	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
	Minor-1	4	20	5	5	5	5	-	-	5	5	5	5	-	-	5	5	5	5	-	-
MIL/AEC (First Language)	English	4	20	5	5	5	5	-	-	5	5	5	5	-	-	5	5	5	5	-	-
Second Language (Telugu, Hindi, Urdu, etc.)		4	20	5	5	5	5	-	-	5	5	5	5	-	-	5	5	5	5	-	-
Multi- Disciplin ary Course	MDC 1	1	4	-	-	-	-	4	-	-	-	-	-	4	-	-	-	-	-	4	-
Sec 1, 2		2	4					2	2					2	2					2	2
Sec 3, 4		2	4					2	2					2	2					2	2
Value added course (VAC)	VAC 1, 2	2	6	-	-	-	-	3	3	-	-	-	-	3	3	-	-	-	-	3	3
Internshi ps	Interns hip / Project	1	4	-	-	-	-	-	4	-	-	-	-	-	4	-	-	-	-	-	4
Total Credits in each semester			142	2 5	2 5	2 5	2 5	2 1	2 1	2 5	2 5	2 5	2 5	2 1	2 1	2 5	2 5	2 5	2 5	2 1	2 1
Total Credits in UG				142						142						142					
Credits under Non- CGPA (Community engagement and service)		NSS /NCC /sport s/ Extra curric ular	6	Upto 6 (2 in each year)						Upto 6 (2 in each year)						Upto 6 (2 in each year)					
		IKS	4	Upto 4 (2 in each, after I & II years)						Upto 4 (2 in each, after I & II years)						Upto 4 (2 in each, after I & II years)					

PAPER – V:: (A) MODERN PHYSICS

(DSE-1: ELECTIVE)

Theory:	4 Hours/Week;	Credits: 4	Marks: 100 (Internal: 20; External: 80)
Practical:	3 Hours/Week	Credits: 1	Marks: 25

UNIT - I

SPECTROSCOPY

Atomic Spectra: Introduction - Drawbacks of Bohr's atomic model – Sommerfeld's elliptical orbits -relativistic correction (no derivation). Stern & Gerlach experiment, Vector atom model and quantum numbers associated with it. L-S and j-j coupling schemes. Spectral terms, selection rules, intensity rules – spectra of alkali atoms, doublet fine structure, Zeeman Effect, Paschen-Back Effect and Stark Effect (basic idea).

Molecular Spectroscopy: Types of molecular spectra, pure rotational energies and spectrum of diatomic molecule. Determination of inter nuclear distance. Vibrational energies and spectrum of diatomic molecule. Raman effect, classical theory of Raman effect. Experimental arrangement for Raman effect and its applications.

UNIT – II

Quantum Mechanics

Inadequacy of classical Physics: Spectral radiation - Planck's law (only discussion). Photoelectric effect - Einstein's photoelectric equation. Compton's effect - experimental verification.

Matter waves & Uncertainty principle: de Broglie's hypothesis - wavelength of matter waves, properties of matter waves. Phase and group velocities. Davisson and Germer experiment. Double slit experiment. Standing de Broglie waves of electron in Bohr orbits. Heisenberg's uncertainty principle for position and momentum (x and p_x), Energy and time (E and t). Gamma ray microscope. Diffraction by a single slit. Position of electron in a Bohr orbit. Complementary principle of Bohr.

Schrodinger Wave Equation

Schrodinger time independent and time dependent wave equations. Wave function properties - Significance. Basic postulates of quantum mechanics. Operators, eigen functions and eigen values, expectation values.

UNIT - III

Nuclear Physics

Nuclear Structure: Basic properties of nucleus - size, charge, mass, spin, magnetic dipole moment and electric quadrupole moment. Binding energy of nucleus, deuteron binding energy, p-p, n-n, and n-p scattering (concepts), nuclear forces. Nuclear models - liquid drop model, shell model.

Alpha and Beta Decays: Range of alpha particles, Geiger – Nuttal law, Gammow's theory of alpha decay. Geiger – Nuttal law from Gammow's theory. Beta spectrum - neutrino hypothesis.

Particle Detectors: GM counter, proportional counter, scintillation counter.

UNIT:IV

Solid State Physics & Crystallography

Crystal Structure: Crystalline nature of matter, Crystal lattice, Unit Cell, Elements of symmetry. Crystal systems, Bravais lattices. Miller indices. Simple crystal structures (S.C., BCC, FCC, CsCl, NaCl, diamond and ZincBlende)


 Mrs. G. Manjula, Chairperson, BoS


 Prof. B. Venkatram Reddy, HoD

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X-ray Diffraction: Diffraction of X-rays by crystals, Bragg's law, Experimental techniques - Laue's method and powder method.

Bonding in Crystals: Types of bonding in crystals - characteristics of crystals with different bondings. Lattice energy of ionic crystals- determination of Madelung constant for NaCl crystal, Calculation of Born Coefficient and repulsive exponent. Born-Haber cycle.

Suggested books:

1. Modern Physics by G. Aruldas & P.Rajagopal. Eastern Economy Edition.
2. Concepts of Modern Physics by Arthur Beiser. Tata McGraw-Hill Edition.
3. Modern Physics by B. Manjunathan and K. Srinivasan. S. Chand & Co.

PAPER – VI :: (A) ELECTRONICS**(DSE-2: ELECTIVE)**

Theory:	4 Hours/Week;	Credits: 4	Marks: 100 (Internal: 20; External: 80)
Practical:	3 Hours/Week	Credits: 1	Marks: 25

Unit - I

Band theory of P-N junction: Energy band in solids (band theory), valence band, conduction band and forbidden energy gap in solids, insulators, semi conductors and pure or intrinsic semiconductors and impure or extrinsic semi-conductors. N-type semi-conductors, P-type semi-conductors, Fermi level, continuity equation.

Diodes: P-N junction diode, Half-wave, full-wave and bridge rectifier. Zener diode & its characteristics. Zener diode as voltage regulator.

UNIT-II

Bipolar Junction Transistor (BJT) – p-n-p and n-p-n transistors, current components in transistors, CB, CE and CC configurations – transistor as an amplifier -RC coupled amplifier – Frequency response (Qualitative analysis).

Feedback concept & Oscillators: Feedback, General theory of feedback–Concepts of oscillators, Barkhausen's criteria, Phase shift oscillator – Expression for frequency of oscillation.

UNIT-III

Special devices- Construction and Characteristics: Photo diode - Shockley diode -Solar cell, Opto-couplers - Field Effect Transistor (FET) - FET as an Amplifier - Uni Junction Transistor (UJT), UJT as a relaxation oscillator - Silicon controlled rectifier (SCR) - SCR as a switch.

UNIT-IV**Digital Electronics**

Binary number system, conversion of binary to decimal and vice-versa. Binary addition and subtraction (1's and 2's complement methods). Hexadecimal number system. Conversion from binary to hexadecimal and vice-versa, Decimal to hexadecimal and vice-versa.

Logic gates:

OR, AND, NOT gates, truth tables, realization of these gates using discrete components. NAND, NOR as universal gates, Exclusive – OR gate (EX-OR). De Morgan's Laws – Verification.

NOTE: Problems should be solved from every chapter of all units.

Suggested Books:

1. Electronic devices and circuits – Millman and Halkias. *Mc.Graw-Hill Education*.
2. Principles of Electronics by V.K. Mehta – *S. Chand & Co.*
3. Basic Electronics (Solid state) – B. L. Theraja , *S. Chand & Co.*
4. A First Course in Electronics- Anwar A. Khan&Kanchan K. Dey, PHI.
5. Physics of Semiconductor Devices- S. M. Sze
6. Physics of Semiconductors- Streetman.
7. Basic Electronics – Bernod Grob.
8. Basic Electronics for B.Sc (Physics) III Year, 2019, Telugu Academy
9. Digital Principles & Applications – A.P. Malvino and D.P. Leach